



Fig. 1: Timer circuit for mosquito destroyer



Fig. 2: Author's prototype of timer for mosquito destroyer

poriser to reduce the amount of toxic chemical vapours in the inhaling air and thus their ill effects. The circuit turns the device on and off at an inter-

val of 15 minutes without reducing the repellent action on mosquitoes. The circuit (Fig. 1) doesn't use any transformer-based power supply and is therefore compact enough to en-

close inside the switchboard. It is directly powered by 230V AC mains through voltage-limiting resistor R6 (22-kilo-ohm). The low-voltage AC is rectified by diodes D1 and D2, filtered by capacitor C3 and regulated by zener diode ZD1 to provide 12V DC.

The timer action is achieved using 14-state binary counter IC HEF 4060 (IC1). Oscillations of the IC are controlled by C2, R2 and R3 and indicated by the flashing of LED1 connected to Q3 output (pin 7) of IC1.

When the circuit is powered-on

using switch S1, IC1 resets through capacitor C1 and resistor R1 and starts oscillating. After 15 minutes, its Q11 output (pin 1) goes high to trigger triac BT136 (triac1) through resistor R5. Once the triac fires, the neutral line reaches the plug socket through M2 terminal of the triac. The vapouriser connected to the plug socket turns on for 15 minutes and then turns off. This cycle repeats until power switch S1 is turned off. Thus the circuit helps to reduce the amount of chemical vapours in the inhaling air. As the vaporiser works only half of the time, it reduces the power consumption to half, doubling the days of usage.

Assemble the circuit on a general-purpose PCB ensuring adequate spacing between the components. Slewing is necessary for the exposed leads of components, especially triacs and diodes. Enclose the circuit in the switch box and connect the plug socket as shown in the circuit diagram. Fig. 2 shows the author's prototype.

Caution. The circuit carries 230V AC, so take utmost care while testing and fixing it to avoid lethal shock. Assemble it only if you are experienced and competent to handle high voltages. ●

PUSHBUTTON CONTROL FOR SINGLE-PHASE APPLIANCES

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This circuit lets you switch off and switch on a single-phase appliance using two separate push switches. Such an arrangement is common for industrial motors (mostly 3-phase) where an isolation is required between power and control circuits. Personal protection under faulty conditions is ensured if the relay is placed remotely. The circuit also safeguards costly devices against frequent power cuts as the device turns off in the event of power failure and remains off until

it is switched-on again.

The switching-on of the device connected to the relay depends upon the

pickup voltage of the relay. When the mains voltage is very low, the output across capacitor C1 becomes less than

